

Instruction Granularity as Planning Width: Formalizing and Benchmarking for Vision-Language-Action Models

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Abstract

Real-world users provide instructions at varying levels of granularity, yet this remains an uncontrolled and unmeasured variable in Vision-Language-Action (VLA) research, leaving its impact unknown. We formalize granularity as the *planning demand* required to ground language into actions, and adapt *rule width* from width-based planning to quantify this demand. To isolate and evaluate this variable, we introduce Mini-BEHAVIOR-Gran, extending all 20 complex tasks with: (i) a rule-set bank providing instruction variants across up to six granularity levels, and (ii) a dynamic instructor that delivers valid instructions at each step. Our analysis reveals a non-monotonic, U-shaped relationship between granularity and performance, with peaks at fine and coarse extremes. We demonstrate an asymmetry in generalization: models trained on coarse instructions transfer to fine ones, but not vice versa. Furthermore, we diagnose how coarse instructions can induce language-ignoring shortcuts due to high visual predictability, and identify mixed-granularity training regimens that best balance task performance with language grounding. Our findings establish instruction granularity as a fundamental dimension for diagnosing and improving multimodal alignment in VLAs.

1 Introduction

When deploying Vision-Language-Action (VLA) models in the real world, a key challenge arises from natural variation in instruction granularity. Users may provide anything from highly precise commands to vague intentions [Clark, 1996]. Handling instruction granularity is essential for human-agent interaction [Norman, 2002], yet current research faces two interrelated gaps that prevent a systematic answer.

First, at the conceptual levels, the community has not yet reached a consensus on how to formally define or measure instruction granularity. A prevalent practice in hierarchical LLM agents [Yang *et al.*, 2024] and recent VLA work [Myers *et al.*, 2024; Shi *et al.*, 2025; Zhong *et al.*, 2025] is to operationalize granularity as the *level of task decomposition*, i.e., whether an instruction specifies only the final goal or

also enumerates subgoals. While intuitive, *decomposition depth* is not a stable notion of granularity but dependent on the task structure: some tasks are intrinsically difficult yet do not admit meaningful hierarchical breakdowns due to *interleaved subgoals* [Sussman, 1975], whereas even simple tasks can be divided into many steps. In parallel, existing Natural Language Processing (NLP) studies of “granularity” typically focus on linguistic aspects, such as the *level of detail* or *discourse structure* of text [Dong and Lapata, 2018; Huang *et al.*, 2025b], rather than adopting an embodied-agent-centric view to measure an instruction’s *informativeness* in terms of reducing the effort required to generate action plans. We instead depart from these proxies by defining instruction granularity as the *planning demand* imposed by an instruction, i.e., how much internal search an agent must perform to ground the instruction into a concrete, executable plan. This framing explicitly decouples granularity from dedicated choices of task decomposition and yields a directly quantifiable measure within *classical planning* frameworks, which is a better approach for analyzing and benchmarking VLA models.

Second, at the empirical level, variations of language granularity in existing VLA benchmarks is entangled with other factors. For example, benchmarks such as BEHAVIOR-1K, which focus on long-horizon activities, typically use abstract goal descriptions, whereas suites like LIBERO, which emphasize short-horizon robotic manipulation, adopt highly concrete commands [Shridhar *et al.*, 2020a; Li *et al.*, 2022; Liu *et al.*, 2023; Li *et al.*, 2024; Mu *et al.*, 2025]. Therefore, it is difficult to disentangle a model’s failure from either challenging perception, or complex low-level control as opposed to the impact of instruction granularity. In short, the field lacks a benchmark that holds tasks fixed while providing parallel instruction sets with rigorously controlled instruction granularity. To this end, the contributions of our paper are:

(1) A Domain-level Formalization: We formalize instruction granularity as *rule width*, a concept adapted from *width-based planning* [Bonet *et al.*, 2019; Drexler *et al.*, 2024]. Intuitively, it measures how many state variables (i.e., *features*) of the world state must be maintained **concurrently** when fulfilling an instruction (§ 3). This metric enables consistent comparison across tasks within the same domain, as it is defined over the shared domain-level feature set Φ . More importantly, instruction granularity now reflects the *informativeness*

tiveness of an instruction when an embodied agent attempts to ground it into an action plan.

(2) A Granularity-Controlled Benchmark: We introduce MINI-BEHAVIOR-GRAN, where each task is paired with multiple instruction variants spanning different granularity levels (§ 4). It is the first-of-kind testbed that isolates instruction granularity as an independent variable, so as to enable controlled studies of its effects on VLA agents.

(3) A Granularity-Centric View of Language Grounding: We reveal a non-monotonic, U-shaped relationship between instruction granularity and VLA performance, identifying productive learning regimes at the extremes of Fine and Coarse instructions (§ 5). While prior work attributes “language-ignoring” primarily to statistical biases in training data, we provide a complementary granularity-centric perspective. We argue that instruction granularity is a structural driver of the trade-off between task success and language reliance. Through extensive evaluation across three action-decoding archetypes (Autoregressive, Diffusion, and Parallel), we demonstrate that coarse instructions can incentivize a bypass of linguistic grounding due to their higher predictability from visual state alone, whereas mixed-granularity training, particularly our proposed *Axial-F* setting, best balances overall performance and language grounding.

Instruction granularity shapes the learning dynamics of VLA models trained on multimodal datasets. It introduces a dual risk: overly fine-grained instructions may degrade language into a mere identifier for specific state-action sequences, encouraging memorization rather than generalized understanding; conversely, overly coarse instructions can weaken language-action alignment, encouraging the model to exploit shortcuts in other data modalities—a risk that has been difficult to isolate and measure. Through our formalization and controlled benchmark, we provide the first systematic study of instruction granularity in VLAs, establishing it as a measurable dimension for future research on language grounding and multimodal alignment.

2 Related Work

Instruction Abstraction and Granularity. In this work, *instruction granularity* refers to the extent to which state and action constraints are explicitly specified versus left for the agent to infer when grounding language into actions. This is distinct from, yet often conflated with the concept of *linguistic abstraction*. The term *abstraction* is viewed as *representational compression*—using a compact description to stand for multiple instances or executions. This definition is common in prior work of computational thinking [Wing, 2010; Lachmy *et al.*, 2022; Zheng *et al.*, 2024]. Importantly, granularity can vary independently of abstraction: “tidy the room” is coarse but non-abstract (it leaves the exact required actions underspecified, yet it lacks representational compression.), whereas “repeat wiping until clean” is relatively specific yet abstract (it prescribes an action but uses a loop condition to compress multiple executions).

While not framed as the term “granularity”, we acknowledge that a rich body of work has focused on augmenting language specificity and its impact, for example through

State Progression	Concurrent Features Tracked (Sequence of Tuples)	Width
Initial State	$\{\langle \rangle\}$	0
Navigate to rag	$\{\langle p_r = X \rangle, \dots, \langle p_r = 0 \rangle\}$	1
Pick up rag	$\{\langle p_r = 0 \rangle, \langle H_r \rangle\}$	1
Navigate to sink	$\{\langle H_r, p_s = X \rangle, \dots, \langle H_r, p_s = 0 \rangle\}$	2
Turn on sink	$\{\langle H_r, p_s = 0 \rangle, \langle T_s \rangle\}$	2
Soak rag	$\{\langle H_r, T_s, p_s = 0 \rangle, \langle S_r \rangle\}$	3
Navigate to Shoe 1	$\{\langle H_r, S_r, p_{f1} = X \rangle, \dots, \langle H_r, S_r, p_{f1} = 0 \rangle\}$	3
Clean Shoe 1	$\{\langle H_r, S_r, p_{f1} = 0 \rangle, \langle C_1 \rangle\}$	3
Navigate to Shoe 2	$\{\langle H_r, S_r, C_1, p_{f2} = X \rangle, \dots\}$	4
Clean Shoe 2	$\{\langle H_r, S_r, C_1, p_{f2} = 0 \rangle, \langle C_1, C_2 \rangle\}$	4
Put rag on floor	$\{\langle C_1, C_2, H_r \rangle, \dots, \langle C_1, C_2, \text{onfloor}_r \rangle\}$	3

Table 1: State progression and width for the “Cleaning k Shoes” ($k = 2$) w/o instructions. Width scales as $w = k + 2$.

coarse-to-fine semantic parsing [Dong and Lapata, 2018; Li *et al.*, 2020]. Other work generates compositional instructions via constraint dropout and sub-task recombination [Huang *et al.*, 2025b], or construct counterfactual instructions to augment data diversity [Glossop *et al.*, 2025].

In embodied instruction following, varying instruction detail is often handled via a simple two-part decomposition: high-level plans are converted into sequences of low-level steps before being grounded into actions [Yang *et al.*, 2024; Shi *et al.*, 2025]. Most recent VLA works, however, operate with static language instructions [Ma *et al.*, 2025; Li *et al.*, 2025]. Some models like InstructVLA and ThinkAct explores having chain-of-thought reasoning steps before action decoding [Yang *et al.*, 2025; Huang *et al.*, 2025a]. Despite this, quantitatively controlled studies on multi-level instruction granularity remain scarce in embodied AI domain.

Language Sensitivity vs. Ignoring in VLA. Recent studies present two seemingly contradictory findings. On one hand, VLA models appear sensitive to language variations, where semantically equivalent rephrasing of seen instructions leads to performance drops [Akula *et al.*, 2022; Karnik *et al.*, 2025]; this suggests a reliance on superficial word patterns. On the other hand, recent analyses identify a “language-ignoring” behavior, where models over-rely on visual context and bypass instructions entirely [Xing *et al.*, 2025; Fei *et al.*, 2025; Xu *et al.*, 2025]. We provide a comprehensive granularity-centric view that reconciles these findings: instruction granularity modulates the balance between language sensitivity and ignoring, with overly coarse instructions leading to language-ignoring shortcuts.

3 Rule Width as Instruction Granularity

3.1 Running Example

We first illustrate *planning width* computation using a running example. Table 1 shows the desired state progressions of a task where two shoes are cleaned using a rag, and the rag is placed on the floor afterwards. The key **environment dynamics** are: (1) the rag must be soaked before it can clean shoes, and (2) the sink must be turned on to soak the rag. We represent states using a set of task-relevant feature variables (Φ): (i) boolean features H_r (holding rag), T_s (sink is turned on), S_r (rag is soaked), C_i (shoe i is cleaned), onfloor_r (rag is on the floor); and (ii) spatial features p_r, p_s, p_f (distance to the rag, sink, and nearest uncleaned shoe, respectively).

First, approaching the rag is a navigation-only segment: the desired state progression is to decrease the distance feature p_r until $p_r = 0$. Thus, the width is 1 as only a single feature needs to be tracked. In this environment dynamics, $p_r = 0$ enables the state transition to H_r . Thus, the rag can be obtained at $p_r = 0$ and the width remains 1. Subsequently, H_r must be maintained while progressing to next states, resulting in a width of 2. The rag becoming soaked requires simultaneous maintenance of three features, i.e., at the sink ($p_s = 0$), holding the rag (H_r), and having the sink turned on (T_s), which yields a width of 3. After soaking, C_1 can be obtained by tracking H_r , S_r , and p_f , giving width 3. The subtlety arises when cleaning the second shoe: one has to track the same features as before, plus maintaining the cleanliness of the first shoe (C_1), which increases the width to 4.

Why no action model? Agents’ action sets constrain how state transitions are executed, but *width* focuses on state-level feature dependencies induced by the environment dynamics. Different agents (e.g., a rover vs. a drone) may have different action spaces to achieve the same feature change, so emphasizing state progressions promotes generalization of width-based concepts across different embodiments.

The impact of *instruction granularity* on VLA behavior remains underexplored largely because the field lacks a **formal, quantifiable, cross-task** measure of granularity. The most common heuristic, *decomposition depth* [Myers et al., 2024; Shi et al., 2025], is unstable as discussed in § 1. To bridge this gap, we formalize *instruction granularity* as *planning demand*—the internal search required to ground an instruction into an executable plan. This perspective captures the intuitive difference between fine- and coarse-grained instructions: fine-grained instructions provide a “breadcrumb trail” that narrows the search, whereas coarse instructions leave a much larger solution space to be explored to reach a state that fulfills the instruction.

We operationalize planning demand using **rule width** (w), adapting the notion of *planning width* from classical planning [Lipovetzky and Geffner, 2012] and its extension to *policy sketches* [Bonet and Geffner, 2021; Drexler et al., 2022; Drexler et al., 2024]. We first introduce the underlying task representation, then model instructions as sets of “if-then” rules that decompose tasks, and finally define rule width as a measure of instruction granularity for these rule sets.

3.2 Symbolic Task Representation

While VLAs operate on raw multimodal observations, we adopt a symbolic, relational state abstraction. Many embodied benchmarks provide such structured representations—object properties and relations—that can be expressed as *fluents* to form symbolic state or goal conditions [Shridhar et al., 2020b; Li et al., 2024]. Formally, we consider a classical planning problem $P = \langle \mathcal{F}, \mathcal{A}, I, G \rangle$, where $\mathcal{F}$ is a set of fluents, $\mathcal{A}$ a set of actions, I the initial state, and G the goal.

To bridge raw states and high-level instructions, we define a set of **features** $\Phi = \{\phi_1, \dots, \phi_n\}$, where ϕ_i is a Boolean or integer-valued function over state space S . These features serve as domain-level abstractions that capture task-relevant aspects of the environment. Thus, each s is associated with a **feature valuation vector** $\phi(s) = (\phi_1(s), \dots, \phi_n(s))$.

3.3 Instructions as Rule Sets $\mathcal{R}$

To study instruction granularity in a controlled manner, we formalize each instruction a symbolic interpretation as a set of “if-then” rules $\mathcal{R} = \{r_1, \dots, r_m\}$ defined over the feature space Φ . This rule-based representation does not aim to capture the full expressiveness of natural language; instead, it distills the constraint/goal structure that governs the computational cost incurred by a symbolic search algorithm (i.e., planning demand). Specifically, each r_i is of the form $C_{r_i} \mapsto E_{r_i}$:

- C_{r_i} (the *condition*) is a conjunction of feature constraints. For a Boolean feature p , a constraint is p or $\neg p$; for a numerical feature n , it is $n = 0$ or $n > 0$.
- E_{r_i} (the *effect*) is a conjunction of feature changes. For a Boolean feature p , an effect is p , $\neg p$, or $p?$ (allowing any change); for a numerical feature n , it is $n \downarrow$ (decrease), $n \uparrow$ (increase), or $n?$ (any change).

These forms capture the core semantics of the instructions we study. A rule r_i is **fulfilled** by a state sequence $(s, \dots, s')$ if $\phi(s) \models C_i$ and the pair $(\phi(s), \phi(s'))$ satisfies the changes in E_i (e.g., $\forall (n \downarrow) \in E_{r_i}, \phi_n(s') < \phi_n(s)$) while keeping unmentioned features constant (i.e., $\phi_j(s') = \phi_j(s)$ for $j \notin E_i$).

In our experiments, all natural-language instructions presented to the VLA agent are designed such that their symbolic interpretations, i.e., the rule sets $\mathcal{R}$ over features Φ , are **well-formed** (i.e., goal-separating and terminating) and it is guaranteed that at least one **sound** and **closed** [Bonet and Geffner, 2021] policy exists on all benchmark instances, so rule-following ensures dead-end-free reachability. This rule-based framing explicitly exposes the latent subgoal structure, enabling controlled variation of how much structural guidance the instruction provides.

3.4 Rule Width as a Granularity Measure

We quantify instruction granularity through *rule width* (w), a metric that captures the peak concurrent planning demand required to ground language into action. The intuition is rooted in the mechanism of **Iterated Width (IW)** search [Lipovetzky and Geffner, 2012]. $IW(k)$ is a novelty-guided breadth-first search algorithm that retains a state only if it makes some previously unseen combination of k feature values true. Thus, if a rule r_i has width $w(r_i) = 2$, then reaching its effect E_i from a state s with $\phi(s) \models C_i$ can be fulfilled by tracking at most two features concurrently (to be made precise with a running example in § 3.5); consequently, $IW(2)$ is guaranteed to find a shortest-path plan to fulfill r_i if one exists. $w(r_i) = 0$ is a special case where r_i can be fulfilled in a single step.

The granularity of an instruction (i.e., rule set $\mathcal{R}$) is then defined as the maximum width across its rules:

$$w(\mathcal{R}) = \max_{r_i \in \mathcal{R}} w(r_i). \quad (1)$$

3.5 Running Example Continued

We continue to demonstrate three instruction sets, each representing a different width level, with two variants visualized in Figure 2. We also append a **counting feature** in Φ to track the number of uncleaned shoes, denoted as N_f .

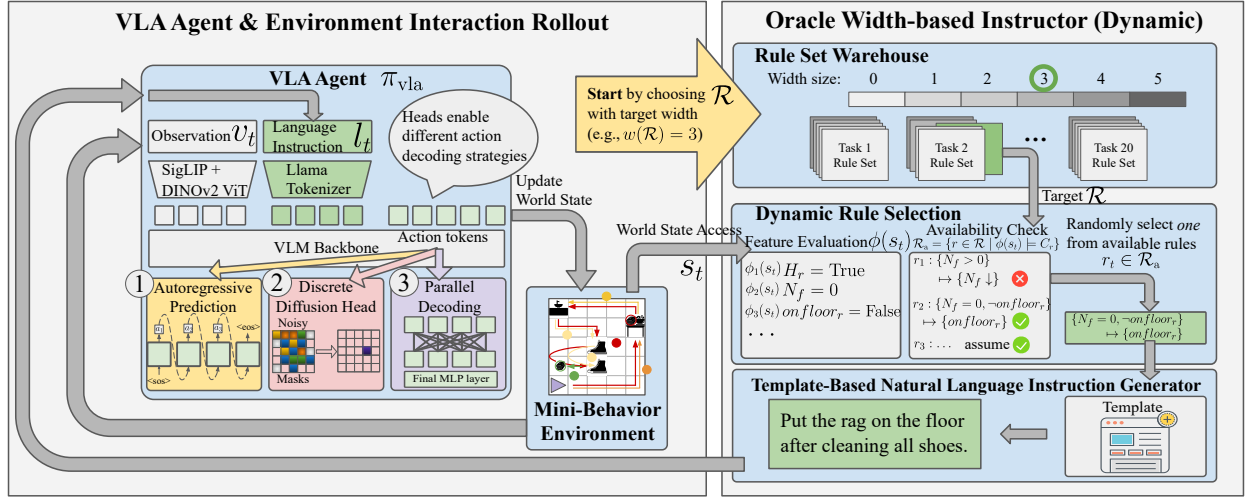


Figure 1: MINI-BEHAVIOR-GRAN Experimental Pipeline. We evaluate the impact of instruction granularity across three VLA action-decoding strategies. During inference time, an Oracle Instructor evaluates the current world state to decide available rule set $\mathcal{R}_a$ and active rule r_t . A rule-based natural language instruction l_t is generated accordingly. The process iterates until completion or max steps.

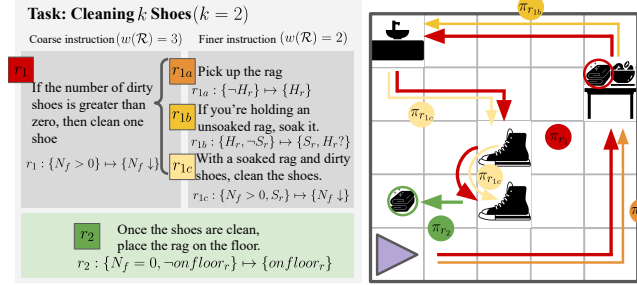


Figure 2: (Left) factoring r_1 into r_{1a} - r_{1c} reduces the planning complexity that the agent must handle internally. More explicit linguistic guidance makes the mapping to action easier to learn. (Right) the long-horizon trajectory for the coarse rule (π_{r_1} , red) is broken into shorter segments by the finer-grained instructions.

Coarse-Grained Instruction ($w(\mathcal{R}) = 3$) This instruction states only the high-level rules and their desired ordering, leaving the internal structure of each sub-task implicit:

- $r_1 : \{N_f > 0\} \mapsto \{N_f \downarrow\}$ (clean a shoe; $w = 3$)
- $r_2 : \{N_f = 0, \neg on_{floor_r}\} \mapsto \{on_{floor_r}\}$ (put the rag on the floor after cleaning all shoes; $w = 1$)

Middle-Grained Instruction ($w(\mathcal{R}) = 2$) This variant decomposes r_1 to explicitly state the *soaking* process:

- $r_{1a} : \{\neg H_r\} \mapsto \{H_r\}$ (pick up the rag; $w = 1$)
- $r_{1b} : \{H_r, \neg S_r\} \mapsto \{S_r, H_r?\}$ (soak the rag; $w = 2$)
- $r_{1c} : \{N_f > 0, S_r\} \mapsto \{N_f \downarrow\}$ (clean; $w = 2$)
- $r_2 : \{N_f = 0, \neg on_{floor_r}\} \mapsto \{on_{floor_r}\}$ (put the rag on the floor; $w = 1$)

This rule set is well-formed: it is goal-terminating via r_2 , and the rules are sequentially reachable, meaning the successor state of any fulfilled rule (that is not the goal) satisfies the condition of at least one rule in $\mathcal{R}$. The set is also robust to

perturbations: if an agent drops the rag after soaking it ($\neg H_r$), the rule set remains applicable by reverting to r_{1a} to reacquire the rag before proceeding to the cleaning task (r_{1c}).

Width Analysis of r_{1b} : From an algorithmic perspective, the width $w(r)$ is the smallest k such that $IW(k)$ solves r . Consider r_{1b} (soaking the rag): if the instruction were expressed as a isolated transition $\{\neg S_r\} \mapsto \{S_r\}$, an agent would need to concurrently coordinate three features: holding the rag (H_r), reaching the sink ($p_s = 0$), and toggling the sink (T_s). However, because r_{1b} 's precondition requires H_r to be established (via r_{1a}), and given that IW search explores states in breadth-first order, any plan that undoes an established feature is necessarily longer than one that preserves it and is subsequently pruned as suboptimal. Hence, the agent can treat H_r as a maintained rather than an actively tracked feature. This reduces the concurrent feature tracking to $\langle T_s, p_s = 0 \rangle$, yielding $w(r_{1b}) = 2$. Although the VLA agent does not perform explicit IW search, this width serves as a rigorous proxy for the inherent planning burden imposed by the instruction.

Fine-grained Instruction ($w = 1$) This variant further decomposes the task into six rules where each transition addresses at most one feature change. The task is fully serialized into three stages: preparation ($r_1 : \{\neg H_r\} \mapsto \{H_r\}$; $r_2 : \{\neg T_s\} \mapsto \{T_s\}$; $r_3 : \{H_r, T_s, \neg S_r\} \mapsto \{S_r\}$); cleaning ($r_4 : \{N_f > 0, \neg H_r, S_r\} \mapsto \{H_r\}$; $r_5 : \{N_f > 0, H_r, S_r\} \mapsto \{N_f \downarrow\}$); and storage ($r_6 : \{N_f = 0, \neg on_{floor_r}\} \mapsto \{on_{floor_r}\}$). The higher rule count and denser preconditions is the explicit indicators of increased instruction granularity. This shift displaces the planning complexity to explicit structural textual guidance. The resulting training data exhibits a more direct language-to-action mapping, thereby reducing the grounding gap that VLA agents must learn to bridge.

To communicate these rules to VLA agents, we translate each rule into a natural-language sentence via a template-based generator. These templates verbalize the features within each rule; e.g., $\{H_r, \neg S_r\} \mapsto \{S_r\}$ is realized

as “If you are holding the rag but the rag is not yet soaked, soak the rag.” This construction preserves the granularity differences defined by the rule width. Additional language prompt examples are provided in § E.

4 Mini-BEHAVIOR-Gran Benchmark

To disentangle the influence of instruction granularity in VLA performance, we seek an environment that (1) retains long-horizon and heterogeneous activities so that instruction guidance meaningfully modulates the planning demand, while (2) simplifies perception and low-level control to avoid confounding factors. As such, we use MINI-BEHAVIOR [Jin *et al.*, 2023] as our testbed. Detailed justification for our choice over other environments is provided in § A.

We introduce MINI-BEHAVIOR-GRAN, the first VLA benchmark specifically designed to enable a controlled study and formal quantification of instruction granularity via *rule width*. Specifically, we define a shared, domain-level feature set Φ that captures task-relevant attributes across all 20 complex household tasks on top of the symbolic state abstraction from the original benchmark. For each task, we design rule sets with varying widths (up to six levels). We verify that all rule sets are well-formed and solvable. We utilize an Oracle Instructor to deliver instructions during inference. This ensures that any observed performance drops are strictly attributable to the model’s inability to ground specific granularity levels, rather than stochastic variations in instruction quality or applicability that would confound our cross-granularity analysis (see Figure 1 right panel). At each time t , it evaluates the current feature vector $\phi(s_t)$, identifies the set of available rules $\mathcal{R}_a = \{r \in \mathcal{R} \mid \phi(s_t) \models C_r\}$. We sample an active rule $r_t \in \mathcal{R}_a$ and it is then processed by a template-based translator to generate a natural language instruction l_t , to match the VLA model’s input format. l_t is paired with the current observation v_t and fed into the VLA model to predict action chunk $(a_t, \dots, a_{t+k}) \sim \pi_{\text{vla}}(\cdot \mid v_t, l_t)$.

5 Experiments

5.1 Experimental Setup

Task Suite & Training. We use MINI-BEHAVIOR-GRAN, which contains 20 complex tasks sharing the same feature set Φ . We further classify instruction granularity into three *relative* classes of *Fine* (F), *Medium* (M), and *Coarse* (C) via per-task quantile binning. In most cases, this mapping corresponds to $w = 0$ for Fine, $w = 1$ for Medium, and $w \geq 2$ for Coarse. We evaluate seven training settings: (1) **Pure F/M/C** ($1 : 0 : 0$): trains exclusively on one granularity class; (2) **Centroid** ($1 : 1 : 1$): uniformly sample from all three classes; (3) **Axial F/M/C** ($3 : 1 : 1$): emphasizes one class with some exposure to the rest classes. Our dataset provides 50 training and 10 evaluation instances per task, each paired with reference trajectories for every granularity level generated by BFWS width-based planner from Lipovetzky and Geffner [2017] (see § D for details).

Action-Decoding Archetypes. The primary architectural difference among latest VLA models lies in their action-decoding strategies [Zhong *et al.*, 2025]. Hence, we employ

a unified VLM backbone but vary three representative action-decoding strategies (see Figure 1 left panel): (1) **Autoregressive** (AR), exemplified by RT-2 [Zitkovich *et al.*, 2023], OpenVLA [Kim *et al.*, 2024] and others [Reed *et al.*, 2022; Huang *et al.*, 2024; O’Neill *et al.*, 2024] that predict actions sequentially via causal attention; (2) **(Discrete) Diffusion**, which uses diffusion process for action chunk prediction [Black *et al.*, 2025; Bjorck *et al.*, 2025; Shukor *et al.*, 2025]. We employ discrete diffusion [Liang *et al.*, 2025] for compatibility with the discrete action space of our testbed; (3) **Parallel Decoding**, utilizing bidirectional attention for single-pass action prediction to improve throughput, as demonstrated by OpenVLA-OFT [Kim *et al.*, 2025], PD-VLA [Song *et al.*, 2025] and other variants [Yin *et al.*, 2025; Wang *et al.*, 2025; Xiao *et al.*, 2025]. Details on model hyperparameters, hardware and training schedules are in § F.

Research Questions. We address the following questions: (Q1) **Granularity Trade-offs.** How does instruction granularity impact Success Rate (SR) on within splits? (Q2) **Generalization.** Can VLA agents trained on certain granularities generalize to other unseen granularities? (Q3) **Failure Dynamics.** When failing, does the agent get stuck (stagnation) or undo achieved progress (regression)?

5.2 Main Results

We first answer (Q1), short answers as follows:

(A1) *VLA performance follows a non-monotonic U-shape, peaking at the Fine and Coarse extremes.*

To establish a performance baseline, we first evaluate the models under matched granularity, where training and testing occur on the same instruction class ($F \rightarrow F$, $M \rightarrow M$, $C \rightarrow C$). As shown in Figure 3, all three action-decoding archetypes demonstrate a non-monotonic U-shaped discovery. Note that all instructions are generated via a unified template-based translator that directly verbalizes symbolic rule sets, ensuring uniform vocabulary, syntax, and phrasing across granularity levels and isolating rule width as the causal factor. We interpret this U-shaped curve as follows. When instructions provide a step-by-step mapping to actions (i.e., $w = 0$), the learning burden is minimal, as the agent can directly follow instructions with little reasoning or planning effort. At the coarse end ($w \geq 2$), agents are motivated to develop internal planning skills to ground coarse instructions into actions. Importantly, since each coarse instruction maps to a broader set of visual and action patterns, it helps develop more robust and generalizable language grounding by providing more diverse training data. In contrast, medium instructions ($w = 1$) occupy a sub-optimal grounding regime. They are too sparse to provide a direct language-to-action mapping, yet not coarse enough to foster internal planning skills, thereby resulting in the performance decline.

From a width-based planning perspective, the U-shaped pattern surprises because coarser instructions should monotonically increase planning demands, making learning progressively harder. Yet, a common confounding factor is instruction sequence length (i.e., the number of rule calls required to complete a task), which also impacts SR due to error accumulation. Note that our evaluation uses the same underlying task set across granularities, and thus the reference

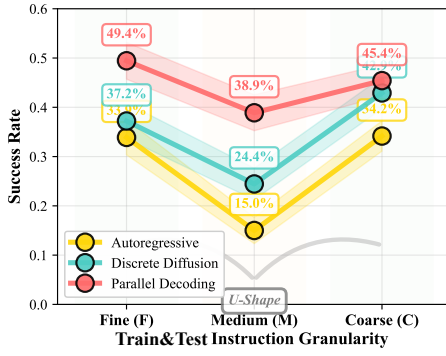


Figure 3: In-Distribution (matched train/test) SR across granularities (i.e., train Fine→test Fine, etc). A U-shaped trend appears.

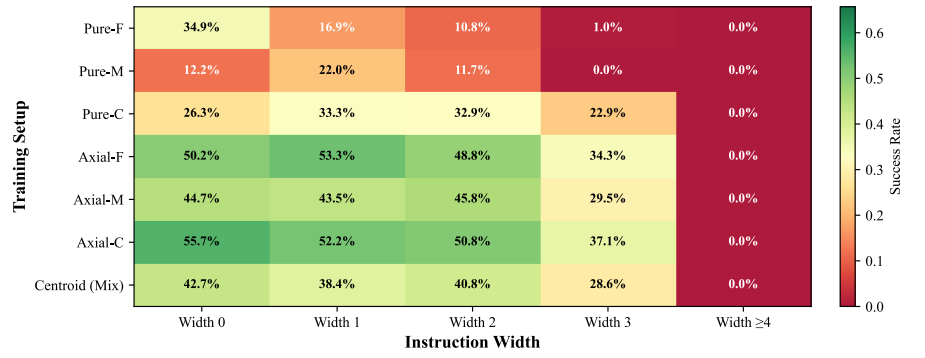


Figure 4: The x-axis are test instruction width, and the y-axis are training setup. Transfer is asymmetric: train coarse → test fine works, but train fine → test coarse does not. Mixed-granularity training improves generalization. *Axial-C* training setting performs best overall.

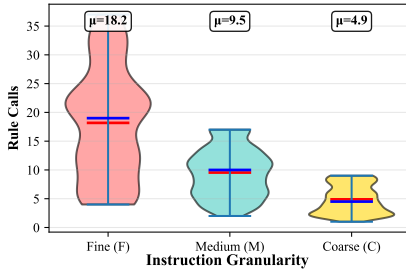


Figure 5: Rule call distribution across instruction granularities. # of rule calls decreases with coarser instructions. It decouples instruction length from width.

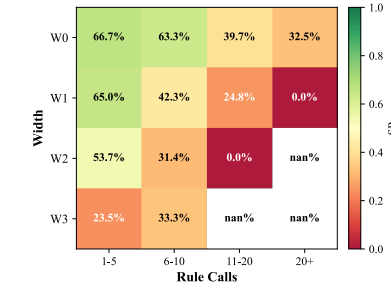


Figure 6: Impact of width and horizon on SR. SR decays with task horizon; larger instruction width imposes an additional burden even when horizon is fixed.

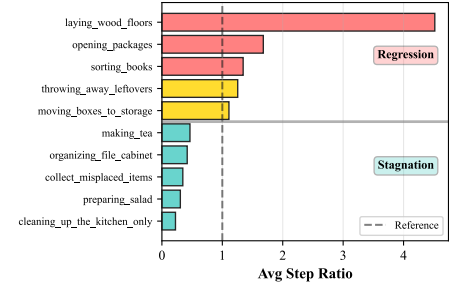


Figure 7: Low-width but long task horizon (e.g., laying woods) tends to cause regression, while high-width tasks tend to stagnate. These contribute to the failure of planning.

atomic action sequence length is held constant. The only entangled factor is thus the rule call count. As shown in Figure 5, coarser instructions require fewer rule calls to accomplish the same task, as each instruction maps to a longer action sequence. The medium granularity operates at a moderate number of rule calls. This decouples instruction sequence length from instruction width and confirms the U-shape cannot be attributed to medium-granularity tasks requiring excessively longer instruction sequences or rule calls.

To further decouple the effects of instruction width and sequence length on VLA performance, we analyze Figure 6, obtained by *Centroid* setting as it has balanced exposure to all granularities. The heatmap provides empirical evidence that, for a fixed number of rule calls, SR decreases as instruction width increases. This demonstrates that the width metric, originally developed for symbolic planning, is a meaningful proxy for the difficulty faced by learning-based VLA agents.

Next, we evaluate cross-granularity generalization, where models are trained on one instruction granularity and evaluated on another. Results are summarized in below:

(A2) *Language grounding is asymmetric: models trained on coarse texts generalize to detailed ones, but not vice versa.*

Figure 4 reveals a clear asymmetry: models trained on coarse instructions generalize well to fine ones, demonstrating robust language grounding that abstracts away from granularity. Conversely, models trained on fine instructions do

not generalize to coarse ones, implying a form of grounding that overfits to step-by-step specification. Mixed-granularity training mitigates this issue, with the coarse-weighted *Axial-C* variant yielding the best generalization overall.

For (Q3), the failure mode is dependent on task structure: (A3) *Low-width but long-horizon tasks tend to cause regression, while high-width tasks tend to stagnate.*

We interpret these failure modes through the lens of width-based planning (see Figure 7). In high-regression tasks, which often require manipulating multiple objects toward a collective goal, agents frequently exhibit subgoal serialization failures. In these settings, the model fails to maintain satisfied fluents over time, cycling back to earlier states. Conversely, stagnation in high-width tasks (e.g., making tea) reflects a width-planning bottleneck: as simultaneous feature dependencies increase ($w \geq 4$), agents struggle to track multiple simultaneous feature dependencies to enable a valid state progression. More analysis is in § G.

6 Dilemma of Coarse Instruction Training

Our analyses in (Q1) and (Q3) suggest a practical advantage for training with coarse instructions: models maintain strong in-distribution performance while generalizing effectively to unseen, finer-grained instructions. To further evaluate this generalization capability, we test robustness to more extreme instruction variations: providing only a static goal description

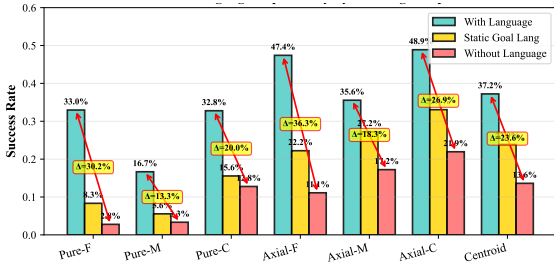


Figure 8: Both mixed and coarse-only training “generalize” better in novel instruction settings (smaller SR gap).

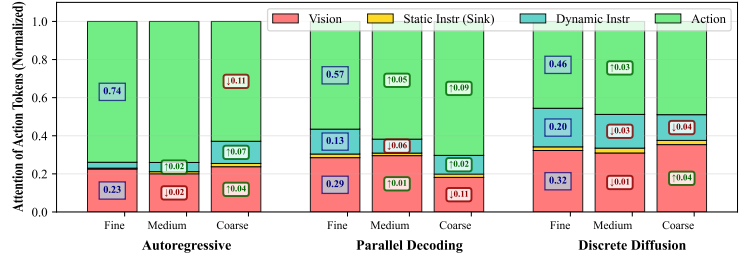


Figure 9: Attention from action tokens. Self-attention dominates, followed by vision; No clear trend across granularity levels.

(fixed for the entire episode, akin to the LIBERO testbed [Liu et al., 2023]) and providing no instruction at all.

Results are shown in Figure 8. Mixed-granularity training, particularly the *Axial-C* variant, achieves the highest SR when presented with the unseen static-goal descriptions. More strikingly, when the language input is entirely absent, *Axial-C* maintains a non-trivial SR of 21%. This raises a fundamental question: does the success of coarse instruction training stem from improved generalization via robust language grounding, or from “language ignoring”, where the policy becomes invariant to language inputs?

Quantifying Language Ignoring. “Language ignoring” is a phenomenon recently identified in VLA training, where agents appear to over-rely on visual patterns, making their policies invariant to language input [Xing et al., 2025; Fei et al., 2025]. A standard metric for this is $\Delta\text{SR} = \text{SR}_{\text{w/ lang}} - \text{SR}_{\text{w/o lang}}$, where a smaller gap indicates stronger ignoring. Our data reveals a trend: models trained purely on coarse instructions (*Pure-C*) exhibit a smaller ΔSR (20.0%) than those trained on fine instructions (*Pure-F*), suggesting coarser training may induce greater language ignoring.

Attention Analysis. To probe the model’s internal reliance on language more directly, we analyze the attention distribution from action tokens to the vision, language, and self-modalities (see Figure 9). Contrary to a simple visual-shortcut hypothesis, we find no clear trend of decreasing attention to the language modality as instruction granularity coarsens. Across all training regimes, self-attention receives the largest share, followed by vision, with language consistently receiving the smallest portion. Notably, autoregressive models attend least to language, which could explain their lower overall SR in Figure 3.

Why Coarse Instructions Encourage Language Ignoring? Xu et al. [2025] posit a causal mechanism: if the language instruction l is highly predictable from visual context v (i.e., large $P(l | v)$), a multimodal model may learn to bypass language processing to save computational energy. This aligns with our benchmark, where expert trajectories are fixed but described at varying granularities: consider a surjective mapping f from a fine-grained label space $L_{\text{fine}} = l_1, \dots, l_n$ to a coarser one $L_{\text{coarse}} = c_1, \dots, c_m$ with $m < n$. The conditional probability for a coarse label aggregates over its constituent fine labels: $P(c_j | v) = \sum_{l_i \in f^{-1}(c_j)} P(l_i | v)$. Since the same total probability mass is distributed over fewer coarse categories, the average $P(c_j | v)$ is higher. According

to the theory, this increased predictability could encourage language ignoring for coarsely-trained models.

Mixed-granularity training mitigates this effect by increasing the diversity of the language space L seen during training, which lowers the average $P(l | v)$. We thus see all mixed-granularity models exhibit larger ΔSR gaps than their pure counterparts, indicating reduced language ignoring. Among these, the fine-weighted mixture *Axial-F* emerges as a particularly balanced choice, achieving a success rate comparable to *Axial-C* while maintaining a substantially larger ΔSR .

Our analysis provides a complementary perspective on the language ignoring phenomenon, demonstrating its variation across instruction granularities, a factor not explored in prior work. The trade-off between final performance and language reliance highlights an open challenge in VLA design: how to structure training for both robust generalization and faithful language grounding.

7 Conclusion and Future Work

This work formalizes and systematically studies how instruction granularity impacts the performance and generalization of VLA agents. Our controlled analysis shows that VLA performance is highly sensitive to granularity, exhibiting a non-monotonic U-shaped relationship that suggests real-world systems may benefit from deliberate instruction granularity design. We provide a granularity-centric diagnosis of the “language ignoring” phenomenon, linking it to instruction predictability and identifying mixed-granularity training as an effective mitigation strategy.

Future Work. Several promising directions follow. First, moving beyond symbolic, grid-based domains to estimate *planning width* in complex, continuous environments would greatly enhance generalizability. This requires developing methods to approximate *width* from ambiguous observations and diverse instruction formats. Second, to enable this, a key challenge is the automatic derivation of relevant feature sets. Future work could develop learnable feature modules that dynamically extract or construct planning-relevant state abstractions directly from multimodal inputs (pixels, knowledge bases, or language descriptions), reducing reliance on hand-designed symbolic predicates and bridging the gap between vectorized data and structured planning models.

Use of LLM Statement An LLM was used only for polish-ing language; the scientific content is entirely original.

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